

ADVANTAGES & DISADVANTAGES

Advantages of UCab

1. User-Friendly Interface

Provides a simple, intuitive, and responsive interface for easy cab booking.

2. Secure Authentication

Uses JWT authentication and encrypted passwords (bcrypt) to ensure secure user login and data protection.

3. Quick Cab Booking

Allows users to book rides efficiently by selecting pickup and destination locations.

4. Fare Estimation

Displays an estimated fare before confirming the ride, helping users make informed decisions.

5. Booking History

Maintains a complete record of previous bookings for easy reference.

6. Responsive Design

Works seamlessly across desktops, laptops, tablets, and mobile devices.

7. Efficient Data Management

MongoDB enables fast and flexible storage and retrieval of user and booking data.

8. Scalable Architecture

The MERN stack architecture supports future enhancements such as GPS tracking, online payments, and driver management.

9. RESTful API Integration

Ensures smooth communication between the frontend and backend using REST APIs.

10. Open-Source Technologies

Built using free and widely adopted technologies, reducing development costs.

Disadvantages of UCab

1. No Real-Time GPS Tracking

The current version does not provide live tracking of drivers and rides.

2. No Online Payment Gateway

Payments are not integrated and must be handled outside the application.

3. Limited User Roles

The current implementation focuses primarily on user functionality and does not include complete driver or admin dashboards.

4. Internet Dependency

Requires a stable internet connection to access all features.

5. Limited Cab Availability

Since the project is a prototype, it does not connect to real cab service providers.

6. Basic Fare Calculation

Fare estimation is based on predefined logic rather than real-time traffic and distance data.

7. No Push Notifications

Users do not receive booking confirmations or ride updates through notifications.

8. Limited Security Features

Advanced security features such as two-factor authentication (2FA) and account recovery are not implemented.

9. No Ride Rating System

Users cannot rate drivers or provide feedback in the current version.

10. Local Deployment

The project is primarily designed for local development and requires additional configuration for cloud deployment.